(1) Serii - Criterii

Criteriul necesar

$\lim_{n\to\infty} x_n \neq 0 \Rightarrow \sum_{n\geq 1} x_n$ divergentă

Exemple fundamentale

(1) Seria geometrică $\sum_{n\geq 0} a\cdot q^n \quad a, q \in \mathbb{R}$

$$S_n = a + aq + \dots + aq^n = \begin{cases} a\cdot\dfrac{1-q^{n+1}}{1-q}, & q \neq 1 \\ (n+1)\cdot a, & q = 1 \end{cases}$$

$|q|<1 \Rightarrow \sum_{n\geq 0} aq^n$ converg.

$|q|\geq 1 \Rightarrow \sum_{n\geq 0} a\cdot q^n$ divergentă

(2) Seria Riemann $\sum_{n\geq 1} \frac{1}{n^\alpha} \quad \alpha \in \mathbb{R}$

$\alpha \leq 0 \Rightarrow \sum_{n\geq 1} \frac{1}{n^\alpha}$ diverg.

$\alpha > 0 \Rightarrow x_n = \frac{1}{x^\alpha}$ șir de nr. pozit. descresc

$$\Rightarrow \sum_{n\geq 1} \frac{2^n}{(2^n)^\alpha} \sim \sum_{n\geq 1} \frac{1}{2^{(\alpha-1)n}}$$

$\Rightarrow$ pt

$\alpha > 1 \quad \sum_{n\geq 1} x_n$ converg.

$\alpha \leq 1 \quad \sum_{n\geq 1} x_n$ diverg.

Criteriul de comparație ($x_n \le y_n$; $x_n \ge 0$; $y_n \ge 0$)

$\sum_{n\ge 1} y_n$ converg $\Rightarrow \sum_{n\ge 1} x_n$ converg.

$\sum_{n\ge 1} x_n$ diverg $\Rightarrow \sum_{n\ge 1} y_n$ diverg.

Criteriul de comp. la limită $\ell = \lim_{n\to\infty} \frac{x_n}{y_n}$

$\ell \in (0; +\infty) \Rightarrow \sum_{n\ge 1} x_n \sim \sum_{n\ge 1} y_n$

$\ell = 0$

$\sum_{n\ge 1} y_n$ conv. $\not\Rightarrow \sum_{n\ge 1} x_n$ conv.

$\ell = \infty$

$\sum_{n\ge 1} y_n$ div. $\not\Rightarrow \sum_{n\ge 1} x_n$ diverg

Criteriul raportului $\ell = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$

$\ell < 1 \Rightarrow \sum a_n$ converg.

$\ell > 1 \Rightarrow \sum a_n$ diverg.

$\ell = 1 \Rightarrow$ nu se știe

Criteriul Rădăcinii $\quad \ell = \lim\limits_{n\to\infty} \sqrt[n]{a_n}$

- $\ell < 1 \Rightarrow \sum a_n$ converg.
- $\ell > 1 \Rightarrow \sum a_n$ diverg.
- $\ell = 1 \Rightarrow$ nu se știe

Criteriul Raabe-Duhamel $\quad \ell = \lim\limits_{n\to\infty} n\left(\frac{a_n}{a_{n+1}} - 1\right)$

- $\ell > 1 \Rightarrow \sum a_n$ converg.
- $\ell < 1 \Rightarrow \sum a_n$ diverg.
- $\ell = 1 \Rightarrow$ nu se știe

Criteriul logaritmic $\quad \ell = \lim\limits_{n\to\infty} \frac{\ln \frac{1}{x_n}}{\ln n}$

- $\ell > 1 \Rightarrow \sum a_n$ converg.
- $\ell < 1 \Rightarrow \sum a_n$ diverg.

Criteriul Abel-Dirichlet $\quad \sum\limits_{n\geq 0} a_n \cdot b_n$

- $a_n \geq 0$
- $a_n \downarrow$
- $\lim\limits_{n\to\infty} a_n = 0$
- $b_1 + b_2 + \ldots + b_n$ mărginit.

$\Rightarrow \sum a_n b_n$ conv.

Criteriul Leibniz

$a_n \geq 0$

$\lim_{n\to\infty} a_n = 0$ $\Rightarrow$ $\sum (-1)^n a_n$ conv.

$a_n \searrow$

Serie absolut convergentă

$\sum_{n\geq 1} |x_n|$ conv $\Rightarrow \sum_{n\geq 1} x_n$ abs. converg.

$\sum_{n\geq 1} x_n$ abs. conv. $\Rightarrow \sum_{n\geq 1} x_n$ conv.

(2) Serii de puteri

Teorema razei de convergență

$$\sum_{n\geq 0} a_n (z-z_0)^n = \sum_{n\geq 0} x_n$$

$$l = \lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

$R = \frac{1}{l}$ R = raza de convergență

$R \in (0;+\infty) \rightarrow |z-z_0| < l \Rightarrow \sum_{n\geq 0} x_n$ A.C.

$|z-z_0| > l \Rightarrow \sum_{n\geq 0} x_n$ D.

$R = 0 \Rightarrow \sum_{n\geq 1} x_n$ Conv.

$R = \infty \Rightarrow \sum_{n\geq 1} x_n$ A.C.

Aflarea lui a_n când știm $f(x) = \sum_{n\geq 1} a_n (x-x_0)^n$

$$a_n = \frac{f^{(n)}(x)}{n!} \quad \forall n \in \mathbb{N}$$

Polinomul Taylor de gradul n

$$T_n(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f^{''}(a)}{2!}(x-a)^2 + \ldots + \frac{f^{(n)}(a)}{n!}(x-a)^n$$

(3) Functii elementare

$$(1+x)^\alpha = 1 + \sum_{n\geq 1} \frac{\alpha(\alpha-1)(\alpha-2)\dots(\alpha-n+1)}{n!} x^n = \sum_{n\geq 0} C_\alpha^n x^n \quad \forall\, |x|<1$$

$$\frac{1}{1+x} = \sum_{n\geq 0} (-1)^n \cdot x^n, \quad |x|<1$$

$$\frac{1}{1-x} = \sum_{n\geq 0} x^n, \quad |x|<1$$

$$\sqrt{1+x} = 1 + \sum_{n\geq 1} \frac{(-1)^{n-1}\cdot 1\cdot 3\cdot\dots\cdot(2n-1)}{2^n\cdot n!}\cdot x^n, \quad |x|<1$$

$$\frac{1}{\sqrt{1+x}} = 1 + \sum_{n\geq 1} (-1)^n \frac{1\cdot 3\cdot\dots\cdot(2n-1)}{2^n\cdot n!} x^n, \quad |x|<1$$

$$e^x = \sum_{n\geq 0} \frac{1}{n!} x^n \qquad \forall\, x\in\mathbb{R}$$

$$\cos x = \sum_{n\geq 0} \frac{(-1)^n}{(2n)!} x^{2n} \qquad \forall\, x\in\mathbb{R}$$

$$\sin x = \sum_{n\geq 0} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \qquad \forall\, x\in\mathbb{R}$$

$$\frac{1}{1+x} = \sum_{n\geq 0} (-1)^n x^n \xrightarrow{\text{int.}} \ln(1+x) = \sum_{n\geq 0} (-1)^n \frac{x^{n+1}}{n+1}$$

$$\ln(1+x) = \sum_{n\geq 0} \frac{(-1)^{n+1}}{n} x^n$$

$x \longrightarrow x^2$ $\quad \frac{1}{1+x^2} = \sum_{n\geq 0} (-1)^n \cdot x^{2n}$

int ↓

$$\operatorname{arctg} x = \sum_{n\geq 0} \frac{(-1)^n}{2n+1} x^{2n+1}$$

$x \longrightarrow -x^2 \Rightarrow \frac{1}{\sqrt{1-x^2}} = 1 + \sum_{n\geq 1} (-1)^n \cdot \frac{1\cdot 3\cdot \ldots \cdot (2n-1)}{2^n \cdot n!} (-x^2)^n$

int. ↓

$$\operatorname{arcsin} x = x + \sum_{n\geq 1} \frac{1\cdot 3\cdot \ldots \cdot (2n-1)}{2^n \cdot n!} \cdot \frac{x^{2n+1}}{2n+1}$$

(4) Functii de mai multe variabile

Diferentiala

$$f(x)-f(a) \simeq Df(a)(x-a)$$

$$Df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

$$Df(a,b,c) = J(a,b,c)\cdot\begin{pmatrix} x \\ y \\ z \end{pmatrix};\quad J(a,b,c) = \begin{pmatrix} \frac{\partial f_1}{\partial x}(a) & \frac{\partial f_1}{\partial y}(b) \dots \\ \frac{\partial f_2}{\partial x}(a) & \dots\dots \end{pmatrix}$$

Diferentiala a doua

$$D^2f(a)(x_1 \dots x_n) = \sum_{i,j=1}^{n} \frac{\partial^2 f}{x_i x_j}(a)\, x_i x_j$$

$$D^2f = \frac{\partial^2 f}{\partial x^2}dx^2 + 2\frac{\partial^2 f}{\partial x \partial y}dxdy + \frac{\partial^2 f}{\partial y^2}dy^2$$

Laplaciaul functiei

$$\nabla f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}$$

Matricea Hessiana

$$H_f(a) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(a)\right)_{i,j=\overline{1,n}}$$

Polinomul Taylor

$$T_2(x) = \underbrace{f(a) + \frac{1}{1!}\left[\frac{\partial f}{\partial x_1}(a)(x_1-a_1) + \frac{\partial f}{\partial x_2}(a)(x_2-a_2) + \dots\right]}_{T_1(x)} +$$

$$\frac{1}{2!}\left[\frac{\partial^2 f}{\partial x_1^2}(a)(x_1-a_1)^2 + 2\frac{\partial^2 f}{\partial x_1 x_2}(a)(x_1-a)(x_2-a) + \dots + \frac{\partial^2 f}{\partial x_4^2}(a)(x_4-a_4)^2\right]$$

Extremele funcției

→ calculăm deriv. part. de ord. I (pct. crit)

→ rezolvăm sistem. $\frac{\partial f}{\partial x_i} = 0$

→ calculăm $\frac{\partial^2 f}{\partial x_i \partial x_j}$ (deriv. part. de ord II)

→ calculăm matricea Hessiană în pct. crit

→ calculăm val. proprii ale mat. Hessiene

Dacă $(-1)^i \Delta_i > 0 \Rightarrow$ pct. e de maxim

Dacă $\Delta_i > 0 \Rightarrow$ pct. e de minim

Dacă toate val. prop. $> 0 \Rightarrow$ minim

Dacă toate val. prop. $< 0 \Rightarrow$ maxim

$d^2 f(a) > 0 \Rightarrow$ minim

$d^2 f(a) < 0 \Rightarrow$ maxim

Se studiază semnul diferenței $f(x,y) - f(a,b)$

dacă $> 0 \Rightarrow$ pct. minim

$< 0 \Rightarrow$ pct. maxim

Extreme cu legături

→ în interior — se face normal (sist. pct. critice)
se verifică dacă pct. crit. respectă condițiile

→ pe frontieră — $F(x,y,\lambda) = f(x,y) + \lambda g(x,y)$
$g(x,y)$ condiția
sistem. $\begin{cases} \frac{\partial F}{\partial x} = 0 \\ \frac{\partial F}{\partial y} = 0 \\ g(x,y) = 0 \end{cases}$

→ în orice punct găsit facem $f(\text{punct})$

(5) Integrale criterii de convergență

Th. Leibniz-Newton

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Integrală improprie – exemple fundam.

$$\int_a^{\infty} \frac{1}{x^\alpha}\,dx \text{ conv.} \iff \alpha > 1$$

$$\int_a^b \frac{1}{(b-x)^\alpha}\,dx \text{ conv.} \iff \alpha < 1$$

Criterii convergență

$$l = \lim_{x \to \infty} x^\alpha f(x)$$

$$l \in (0; +\infty) \Rightarrow \int_a^{\infty} f(x) \sim \int_a^{\infty} \frac{1}{x^\alpha}$$

$$l = \lim_{x \to b} (b-x)^\alpha f(x)$$

$$l \in (0; +\infty) \Rightarrow \int_a^{b} f(x) \sim \int_a^{b} \frac{1}{(b-x)^\alpha}$$

Crit Abel-Dirichlet

$\int_a^\infty f(x)g(x)dx$

f, g cont.

$\lim_{x\to\infty} f(x) = 0$

g are o primitivă mărg.

$\int_a^\infty |f'(x)|\,dx$ conv.

$\Rightarrow \int_a^\infty f(x)g(x)$ conv.

Integ. cu param.

$$F(y) = \int_a^b f(x,y)\,dx \longrightarrow F'(y) = \int_a^b \frac{df}{dy}(x,y)\,dx$$

(6) Integrale Gama și Beta

Integ. Gama (Γ)

$$\Gamma(\alpha)=\int_0^{\infty} x^{\alpha-1}e^{-x}dx$$

$\Gamma:(0,\infty)\longrightarrow\mathbb{R}$, $\alpha>0$

$$\Gamma\left(\frac{1}{2}\right)=\sqrt{\pi}$$

$$\Gamma(1)=1$$

$$\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$$

$$\Gamma(n)=(n-1)!$$

$$\Gamma(\alpha)\Gamma(1-\alpha)=\frac{\pi}{\sin(\alpha\pi)}$$

$$\int_{-\infty}^{\infty} e^{-y^2}=\sqrt{\pi}$$

Integ. Beta (B)

$$B(p,q)=\int_0^1 x^{p-1}(1-x)^{q-1}dx$$

$B:(0,\infty)\times(0,\infty)\longrightarrow\mathbb{R}$

$$B(p,q)=\int_0^1 \frac{y^{p-1}}{(1+y)^{p+q}}dy$$

$$B(p,q)=\frac{\Gamma(p)\Gamma(q)}{\Gamma(p+q)}$$

(7) Integrale Curbilinii

→ integrale definite pe drepte de lungime 1
(au nevoie de o singură variabilă pt. a fi definite)

dreapta: $\begin{cases} x = x_0 + at \\ y = y_0 + at \\ z = z_0 + pt \end{cases}$ $t \in \mathbb{R}$

cerc: $\begin{cases} x = x_0 + R\cos t \\ y = y_0 + R\sin t \end{cases}$ $t \in [0, 2\pi]$

cilindru: $\begin{cases} x = R\cos t \\ y = R\sin t \\ z = at \end{cases}$

Integ. curb. de spețа I

$$\ell(\gamma) = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\, dt$$

lungimea

$$\text{masa}(k) = \int_\gamma \rho\, ds = \int_a^b \rho \cdot \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\, dt$$

masa cu ρ = densitate

$$\int_\gamma f(x,y,z)\, ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{(x'(t))^2 + (y'(t))^2 + (z'(t))^2}\, dt$$

Integ. curb. de spețe a II a

$$C = \int_{\gamma} \bar{v}\, d\bar{r} = \int_a^b P\,dx + Q\,dy + R\,dz$$

Circulația (Lucru mecanic)

$$\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$$
$$d\bar{r} = dx\bar{i} + dy\bar{j} + dz\bar{k}$$

$$\int_{\gamma} P\,dx + Q\,dy + R\,dz = \int_a^b P \cdot x'(t) + Q \cdot y'(t) + R \cdot z'(t)\,dt$$

$\frac{\partial P_i}{\partial x_j} = \frac{\partial P_j}{\partial x_i}$ (cond. necesară)

$\times$ mulțime stelată

$\not\Rightarrow$ câmpul vectorial format din P_{ij} pe mulțimea stelată e câmp de gradienți

$\bar{v}$ câmp de gradienți $\Leftrightarrow \int_{\gamma} \bar{v}\, d\bar{r} = 0$ $\forall$ γ curbă închisă

(8) Integrale duble

I $D = [a,b] \times [c,d]$

$$\iint_D f\,dx\,dy = \int_a^b \left(\int_c^d f\,dy \right) dx = \int_c^d \left(\int_a^b f\,dx \right) dy$$

II $f, g : [a,b] \to \mathbb{R}$ $f \le g$

$D = \{ f(x) \le y \le g(x) \}$ $\iint_D h\,dx\,dy = \int_a^b \int_f^g h\,dy\,dx$

III Coordonate polare $x^2 + y^2 \le R^2$

$$\begin{cases} x = \rho \cos\theta \\ y = \rho \operatorname{sin}\theta \end{cases}$$

$\det J = \rho$ $x^2 + y^2 = \rho^2$

$\rho \in [0,R]$ $\theta \in [0, 2\pi]$

$(x-x_0)^2 + (y-y_0)^2 \le R^2 \longrightarrow \begin{cases} x = x_0 + \rho\cos\theta \\ y = y_0 + \rho \operatorname{sin}\theta \end{cases}$

Coordonate polare generalizate

$\frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1 \longrightarrow \begin{cases} x = a\rho\cos\theta \\ y = b\rho \operatorname{sin}\theta \end{cases}$ $\rho \in [0,1]$ $\theta \in [0, 2\pi)$

$|J| = ab\rho$

(9) Integrale triple

I $\mathcal{R} = [a_1, b_1] \times [a_2, b_2] \times [a_3, b_3]$

x, y fixi $\quad z = f(x, y)$

$$\iiint_{\mathcal{R}} f \, dx\,dy = \int_{a_1}^{b_1} dx \int_{a_2}^{b_2} dy \int_{a_3}^{b_3} f \, dz$$

II multime proiectabila pe xOy

$f, g : D \longrightarrow \mathbb{R} \quad f \leq g$

$$\mathcal{R} = \{ f(x,y) \leq z \leq g(x,y) \}$$

$$\iiint_{\mathcal{R}} h \, dx\,dy\,dz = \iint_{D} \left(\int_{f}^{g} h \, dz \right) dx\,dy$$

III Coordonate cilindrice

$$\begin{cases} x = \rho \cos\theta \\ y = \rho \sin\theta \\ z = z \end{cases} \qquad |J| = \rho$$

Coordonate sferice

$$\begin{cases} x = \rho \sin\theta \cos\varphi \\ y = \rho \sin\theta \sin\varphi \\ z = \rho \cos\theta \end{cases}$$

$\det J = \rho^2 \sin\theta$

$\varphi \in [0, 2\pi] \quad \theta \in [0, \pi]$

Coordonate sferice generalizate

$\mathcal{E}: \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \leq 1$

$\det J = abc\rho^2 \sin\theta$

$$\begin{cases} x = a\rho\sin\theta\cos\varphi \\ y = b\rho\sin\theta\sin\varphi \\ z = \rho\cos\theta \end{cases}$$

Volumul

$$\text{Vol}(\mathcal{E}) = \iiint_{\mathcal{E}} dx\,dy\,dz$$

(10) Integrala de suprafață

param. carteziană $\longrightarrow \begin{cases} x = x \\ y = y \\ z = f(x,y) \end{cases}$

param canonică sferă $\longrightarrow \begin{cases} x = R\sin\theta\cos\varphi \\ y = R\sin\theta\sin\varphi \\ z = R\cos\theta \end{cases}$

$\|\bar{N}(\theta,\varphi)\| = R^2\sin\theta$

Vect. tang. la suprafață

Σ = suprafață

$\Psi(u,v)$ = pct.

$$\frac{\partial\Psi}{\partial u} = \left(\frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u}\right)$$

$$\frac{\partial\Psi}{\partial v} = \left(\frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v}\right)$$

Normala la suprafață

$$\bar{N} = \frac{\partial\Psi}{\partial u} \times \frac{\partial\Psi}{\partial v}$$

$$\bar{v} = \frac{\bar{N}(u,v)}{\|\bar{N}(u,v)\|} \quad \text{versor}$$

Integ. de sup. de speța I

$$\text{Aria}(\Sigma) = \int_\Sigma d\sigma = \iint_D \sqrt{1 + \left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}\, dx\,dy$$

$z = f(x,y)$

$D = pr_{xOy}\Sigma$

$$\int_\Sigma F\,d\sigma = \iint_D F(\Psi(u,v))\,\|\bar{N}(u,v)\|\,du\,dv$$

Integ. de sup. de spețe a II a

$$\mathcal{F}_\Sigma(\bar{v}) = \int_\Sigma \bar{v}\cdot\bar{N}\, d\bar{r}$$

Fluxul

$$\bar{r} = x\bar{i} + y\bar{j} + f(x,y)\bar{k} \qquad z = f(x,y)$$

$$\bar{N} = \pm \frac{\bar{r}_x \times \bar{r}_y}{\|\bar{r}_x \times \bar{r}_y\|} \quad \text{în fc. de sens}$$

$$\mathcal{F}_\Sigma(\bar{v}) = \pm \iint_D \bar{v}\cdot \bar{r}_x \times \bar{r}_y \, dx\,dy$$

↳ determinant

elem. $\bar{v}$
elem. r_x
elem. r_y

$$\mathcal{F}_{\Sigma\cup\Sigma_1}(\bar{v}) = \mathcal{F}_\Sigma(\bar{v}) + \mathcal{F}_{\Sigma_1}(\bar{v})$$

(11) Green - Riemann

→ orientare pe ∂K "în stânga"

→ originea **NU** aparține domeniului

$$C = \int_{\partial K} P\,dx + Q\,dy = \int_{\gamma} \bar{v}\,d\bar{r} = \iint_{K} \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\,dx\,dy$$

$$\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y} \Rightarrow \int_{\partial K} P\,dx + Q\,dy = 0$$

$$\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = C \Rightarrow \int_{\partial K} P\,dx + Q\,dy = C \cdot \text{aria}(K)$$

(12) Gauss - Ostrogradski

divergența

$$\operatorname{div} \bar{v} = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}$$

$$\mathcal{F}_{\Sigma}(\bar{v}) = \int_{\partial K} \bar{v}\,\bar{N}\,d\sigma = \iiint_{K} \operatorname{div}(\bar{v})\,dx\,dy\,dz$$

$$\mathcal{F}_{\Sigma \cup \Sigma_1}(\bar{v}) = \iiint_{K} \operatorname{div}(\bar{v})\,dx\,dy\,dz$$

(13) Formula lui Stokes

$$C = \int_{\gamma} \bar{v}\, d\bar{r} = \int_{\Sigma} \operatorname{rot} \bar{v} \cdot \bar{N}\, d\sigma$$

Circulația

$z = f(x,y)$

$D = pr_{xOy}\, \Sigma$

$$C = \iint_{D} \operatorname{rot} \bar{v} \cdot r_x \times r_y \, dx\, dy$$

$$\mathcal{V}_{\Sigma}(\operatorname{rot}(\bar{v})) = \int_{\Sigma} \operatorname{rot}(\bar{v}) \cdot \bar{N}\, d\sigma \overset{St.}{=\!=} \int_{\gamma} \bar{v}\, d\bar{r}$$

$$\operatorname{Rot}(\bar{v}) = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

(14) **Contractie**

→ f, f' arăt că are exact o soluție

→ X sp. metric complet

→ zic că x din f și imi fac $g: X \to X$

→ Lagrange $g(x) - g(y) = g'(c)(x-y)$

→ modul + sup $|g'(c)|\,|x-y|$

→ fac $g'(x)$ și iau $h(x) = |g'(x)|$

→ fac $h'(x)$ + aflu sup $h(x)$ pt $[0,1]$

→ g contractie de factor k $\xrightarrow[\text{cond.}]{\text{Th.}}$ $\exists!\, \xi$ a.î. $g(\xi) = \xi$
X sp. metric complet

→ fie $x_0 = 0$ $\qquad |x_n - \xi| \le \frac{k^n}{1-k} |x_1 - x_0|$
$x_{n+1} = g(x_n)$

→ iau n care verifică și am $x_n = g(x_{n-1})$

(15) desene

→ sfera $(x-x_0)^2+(y-y_0)^2+(z-z_0)^2=r^2$

→ elipsoidul $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}-1=0$

→ cilindru $\frac{x^2}{a^2}+\frac{y^2}{b^2}-1=0$

→ paraboloid $\frac{x^2}{a^2}+\frac{y^2}{b^2}=2z$

→ conul $\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c}=0$